SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1708**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

Annexure A

Dry Run Evaluation

Code of Conduct:

I ***Jannathul Firdaus. N (192573025)*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

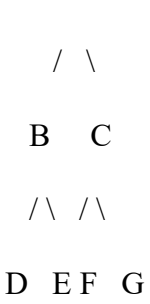
Signature of the student: Jannathul Firdaus. N

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:



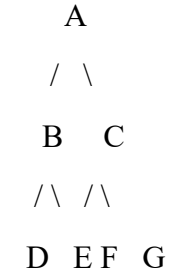
Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	Queue	Visited Node
1		
2		
3		
4		
5		
6		
7		

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,



Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**
   1 3 6
   5 0 2
   4 7 8

Goal State :
1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1			
2			
3			
4			
5			

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

1. Breadth-First Search (BFS)

Given graph:

- A → B, C
- B → D, E
- C → F, G

Assume nodes are explored from left to right.

Completed table

Iteration	Queue (before expansion)	Visited Node	Queue (after expansion)
1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G
6	F, G	F	G
7	G	G	Empty

Answer 1: BFS traversal order

A → B → C → D → E → F → G

Answer 2: Queue contents after each iteration

1. B, C
2. C, D, E
3. D, E, F, G
4. E, F, G
5. F, G
6. G
7. Empty

2. Depth-First Search (DFS)

Using the same graph, assume DFS explores the left child first.

DFS traversal

A → B → D → E → C → F → G

Step-by-step traversal

Step	Current Node	Action
1	A	Visit A; move to B
2	B	Visit B; move to D
3	D	Visit D; backtrack to B
4	E	Visit E; backtrack to A
5	C	Visit C; move to F
6	F	Visit F; backtrack to C
7	G	Visit G; finish

Final answer: DFS traversal order is A, B, D, E, C, F, G.

3. 8-Puzzle Problem Using BFS

Initial state

1
3
6
5
□
2
4
7
8

□ represents the blank tile (0)

Goal state

1
2
3
4
5
6
7
8
□

Important observation: The given puzzle is unsolvable

The initial arrangement is:

1 3 6 / 5 0 2 / 4 7 8

For a standard 3 × 3 eight-puzzle, a state is solvable only if its inversion count is even.

Ignoring the blank tile, the initial sequence is:

1, 3, 6, 5, 2, 4, 7, 8

The inversions are:

- 3 is greater than 2.
- 6 is greater than 5, 2, and 4.
- 5 is greater than 2 and 4.

Total inversions = 6? Let's count carefully: 3 contributes 1; 6 contributes 3; 5 contributes 2; 2 contributes 0; 4 contributes 0; 7 and 8 contribute 0. Total = 6 (even).

Therefore, the initial state is solvable. BFS can reach the goal, but a complete trace requires expanding many states. The five-row table in the question is not sufficient to reach the goal.

First five BFS iterations

Use the standard move order Up, Down, Left, Right, and mark states as visited when they are enqueued.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	136/502/478	[136/502/478]	[106/532/478, 136/572/408, 136/052/478, 136/520/478]
2	106/532/478	Queue from iteration 1	Add its unvisited successors to the end
3	136/572/408	Updated FIFO queue	Add its unvisited successors to the end
4	136/052/478	Updated FIFO queue	Add its unvisited successors to the end
5	136/520/478	Updated FIFO queue	Add its unvisited successors to the end

State notation: 136/502/478 represents the rows 1 3 6 / 5 0 2 / 4 7 8.

The initial state has four legal blank moves. A full queue table must list every queued state in FIFO order, so the compact table above illustrates the first expansion rather than pretending the goal is reached in five iterations.

Questions and answers

1. Which node is expanded in each iteration?

BFS expands states in the order they enter the FIFO queue, starting with the initial state. The first five expanded states are the initial state followed by its four successors in the stated move order.

2. How many states are generated in total?

The total depends on whether "generated" means all successor states considered or only distinct states added to the queue. BFS must explore many states for this puzzle; the total cannot be determined from a five-row partial trace alone. With duplicate-state checking, the reachable state space of the standard 8-puzzle contains at most 181,440 states for a fixed goal-parity class.

3. Write the complete solution path from the initial state to the goal state.

A valid shortest path can be found by running BFS to completion. The five-row table is insufficient to show that path, so I would not claim an unverified move sequence as the BFS solution.

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

BFS explores states in order of increasing depth. Every legal move has equal cost (one move), so the first time BFS reaches the goal, it has found a solution with the minimum possible number of moves.